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Cardiorespiratory Correlation Coefficient Matrices for CNN-Based Sleep Apnea Detection

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***Abstract*— Sleep apnea is a disorder characterized by repetitive temporary cessation of breath, with significant health implications if untreated, thus requiring efficient and accurate identification methods. This paper proposes a novel CNN-based approach for apnea detection from physiological signals: ECG, heart rate, SpO_2, and respiratory signals. The novelty of this work consists in the employment of physiological signal correlation coefficient matrices to represent apnea manifestations. For classification, we employ a pre-trained ResNet-50 model with transfer learning and apply data augmentation to enhance model generalization. Matlab simulation results demonstrate high classification performance, validating the proposed method for sleep apnea detection.**

***Clinical Relevance*— This demonstrated the potential of AI-based physiological signal correlation analysis for decision support system in sleep apnea diagnosis.**

I. Introduction

Sleep Apnea is a disorder characterized by repetitive temporary cessation of breath during sleep, caused either by the obstruction of airways – obstructive sleep apnea (OSA), impaired respiratory control – central sleep apnea (CSA), or both – complex sleep apnea syndrome [1]. These episodes cause oxygen desaturation, poor sleep quality and frequent awakening and, if untreated, can lead to severe health conditions such as hypertension, cardiovascular disease, and metabolic disorders [2-4].

The golden standard for diagnosing OSA is polysomnography (PSG), an overnight monitoring of sleep that records a wide range of physiological signals to evaluate sleep patterns and detect episodes of apnea [5, 6]. The monitored signals include electroencephalogram (EEG), electrocardiogram (ECG), electromyogram (EMG), electrooculogram (EOG), oxygen saturation (SpO_2) and oronasal airflow. Although PSG provides comprehensive sleep assessment, it is expensive, time-consuming and requires specialized equipment. Thus, it limits widespread accessibility for long-term monitoring.

To address these limitations, automated non-invasive apnea detection has gained considerable attention. Machine learning techniques have been proven efficient in biosignal feature extraction and assessment. Bazoukis et al. reviewed the effectiveness of machine learning models in sleep apnea diagnosis, using ECG, pulse oximetry, and acoustic waveform features [7]. Retamales et al. estimate an Apnea-Hypopnea Index (AHI) from abdominal movement, thoracic movement, or pulse oximetry using deep convolutional neural alongside a light-gradient-boost machine, in a scenario which deploys wearable sensors in home environment [8]. Levy et al. proposed a deep learning algorithm to diagnose obstructive sleep apnea from pulse oximetry signals and achieved high accuracy on multiple databases [9]. Salari et al. reviewed machine learning models for sleep apnea detection from ECG characteristics [10].

Convolutional Neural Networks (CNN), a class of deep learning models, have been successfully employed as classifiers. Indeed, CNNs are effective in processing time series, demonstrating a strong potential for processing biomedical data [11].

In this research we propose a novel approach for apnea detection based on PSG signals: ECG, respiratory signals and SpO_2, using CNNs. A key contribution of our work is the development of an original assessment technique, which leverages coefficient correlation matrices to model inter-signal relationships and provide a compact representation of signal interdependencies. Specifically, we compute the Pearson correlation coefficients between pairs of signals to construct these matrices. As such, the correlation coefficient matrices serve as key indicators for abnormal cardiorespiratory coordination, which is the case in apnea. The novelty of our approach is a robust alternative for physiological interpretation through signal correlation patterns. Unlike conventional feature extraction techniques, such as time-domain, spectral analysis and wavelet transform, which aim to capture isolated signal characteristics, the proposed correlation-based representation enables CNNs to learn the spatial patterns representative of physiological dynamics during apnea. This provides an efficient and scalable framework for automated apnea detection.

This paper is organized as follows. Section II presents the proposed correlation-based PSG physiological signal representation technique, aiming for CNN-based apnea classification. Section III presents the CNN implementation for the targeted application. Section IV presents the Matlab results which validate the proposed procedure. Conclusions and future work are finally drawn in section IV.

II. Materials and Methods

In this paper, we propose a novel method to represent the temporal relationships between heart rate, respiratory patterns,

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and oxygen saturation during apnea episodes. This method converts physiological signals into structured images suitable for CNN-based classification.

A. The PhysioNet Apnea-ECG Database

In this work, we use the publicly available PhysioNet Apnea-ECG Database [12], a benchmark dataset for apnea research focused on ECG-based detection of OSA. No new data involving human or animal subjects were collected, and ethical approval was not required. The Apnea ECG Database consists of 70 ECG records from subjects suspected of sleep apnea. Six records, labeled "*er*" and targeted in this work, provide additional respiratory signals: chest wall movement (respC), abdominal wall movement (respA) and nasal airflow (respN), and SpO_2. Additionally, we extract the heart rate (HR) from the ECG based on the duration of the R-R interval.

These signals provide a comprehensive characterization of apnea episodes: the ECG shows potential variability in magnitude, R-R interval and baseline; the heart rate decreases at apnea onset and increases at the resume of breathing; the SpO_2 levels typically decrease during apnea episodes due to interrupted breathing; and the respiratory signals present reduced or absent movement and airflow.

B. Definition of the Correlation Coefficient Matrix

We use the correlation coefficient matrix to identify apnea manifestations in the targeted physiological signals. The correlation coefficient matrix is described as follows.

Consider a signal space expressed generically as

$$S = [sig_1, sig_2, \dots, sig_m] , \tag{1}$$

The correlation coefficient matrix of S is expressed as

$$R = \begin{pmatrix} 1 & \rho_{1,2} & \cdots & \rho_{1,m} \\ \rho_{2,1} & 1 & \cdots & \rho_{2,m} \\ \cdots & \cdots & \cdots & \cdots \\ \rho_{m,1} & \rho_{m,2} & \cdots & 1 \end{pmatrix}, \tag{2}$$

where $\rho_{i,j}$, $i, j = 1 \dots m$, are the Pearson correlation coefficients of signals sig_i and sig_j given by

$$\rho_{i,j} = \frac{\sum_{n=1}^{N}(sig_i(n)-\mu_i)(sig_j(n)-\mu_j)}{\sqrt{\sum_{n=1}^{N}(sig_i(n)-\mu_i)^2 \sum_{n=1}^{N}(sig_j(n)-\mu_j)^2}}, \tag{3}$$

μ_i and μ_j are the means of the two signals respectively, and N is the signal length [13,14].

The Pearson correlation coefficient $\rho_{i,j}$ measures the similarity degree and linear relationship between signals sig_i and sig_j respectively. The coefficient value ranges from -1 to 1, with 1 indicating perfect positive correlation, -1 indicating a perfect negative correlation, and 0 indicating the absence of a linear relationship between the signals.

C. Generated Image Dataset

We use the correlation coefficient matrix to assess the synchronism and trend similarities of the targeted physiological signals: ECG, respiratory signals and SpO_2. The signal spaces for which we evaluate the correlation coefficient matrices, along with the matrix size are listed in Table 1.

The proposed workflow for constructing the signal space follows segmentation of the physiological signals with a frame length corresponding to 200 R beats in the ECG. To balance the number of apnea and normal breathing frames, we have limited the number of segments to 20, an empirical value which creates a satisfactory compromise between processing time for dataset generation and dataset size. Next, 100 shifted versions of each signal frame are generated by applying circular shifting. Considering the symmetric nature of the correlation coefficient matrix, each signal thus contributes 100 x 100 matrix coefficients for autocorrelation and 200 x 200 coefficients for cross-correlation.

TABLE I. CNN PERFORMANCE METRICS

Signal Space	**Correlation Coefficient Matrix**	
	Physiological signals	***Matrix size***
Electrocardiogram	ECG	100x100
Heart rate	HR	100x100
Oxygen saturation	SpO_2	100x100
Cardiorespiratory space	Hr, SpO_2	200x200
Respiratory space	RespC, RespA, RespN	300x300
All signals	Hr, SpO_2, RespC, RespA, RespN	500x500

During normal breathing [1-6], Fig. 1, the physiological signals visualize a regular breathing pattern and cardiac activity. The ECG stays active, determining a stable HR, without significant oscillations or abrupt variations. The chest (RespC) and abdominal (RespA) respiratory signals, synchronized with the nasal airflow (RespN), exhibit a regular breathing pattern, reflecting efficient respiratory activity without significant variations in effort and rhythm. The SpO_2 levels stay high, mostly above 95% indicating proper oxygen exchange without abnormal desaturation events.

In the correlation coefficient matrices, the ECG waveform determines a set of equidistant diagonals corresponding to the periodicity in the R-wave/QRS complex occurrences. Similarly, respiratory signal periodicity, reflecting regular respiratory process, determines multiple narrow diagonals of moderate magnitude. The HR and SpO_2 relative stability, on the other hand, determine gradient patterns without pronounced diagonals.

An apnea episode [1-6] on the other hand is characterized by specific abnormalities in the physiological signals, Fig. 2. The ECG exhibits variations in magnitude, R-R interval and baseline, indicative of changes in autonomic regulation, which translate to non-equidistant diagonals in the correlation coefficient matrix. Chest and abdominal respiratory signals, in correlation with nasal airflow, indicate the cessation of respiratory effort. This is consistent with the observed periodic pattern in HR drops and oxygen desaturation.

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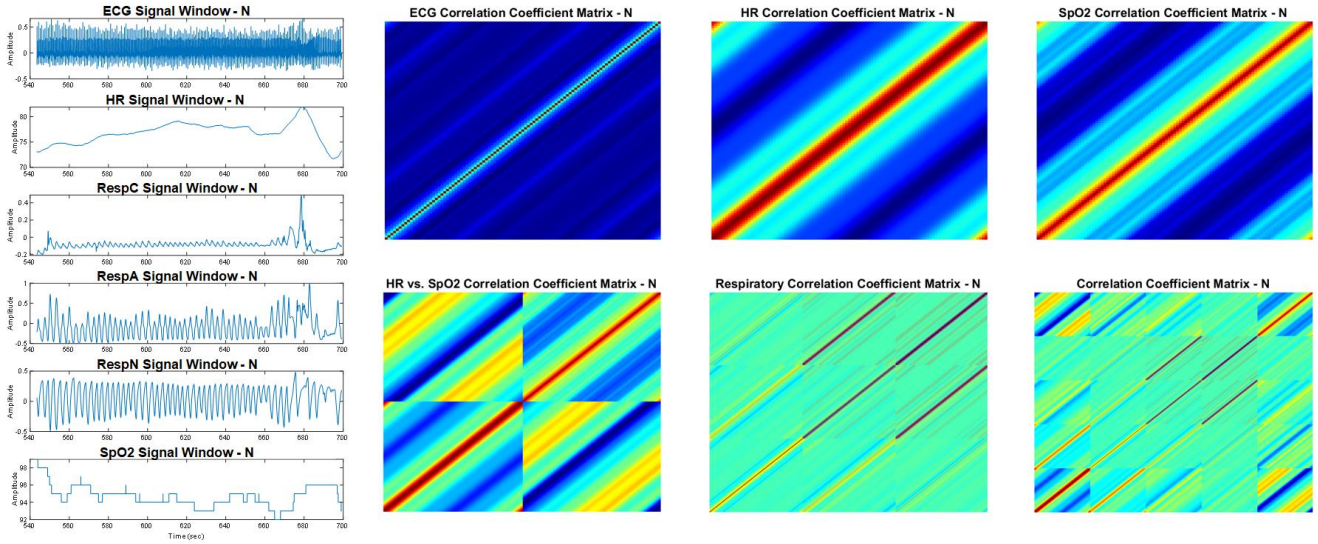


Figure 1. Normal breathing: time-domain signal windows and correlation coefficient matrices.

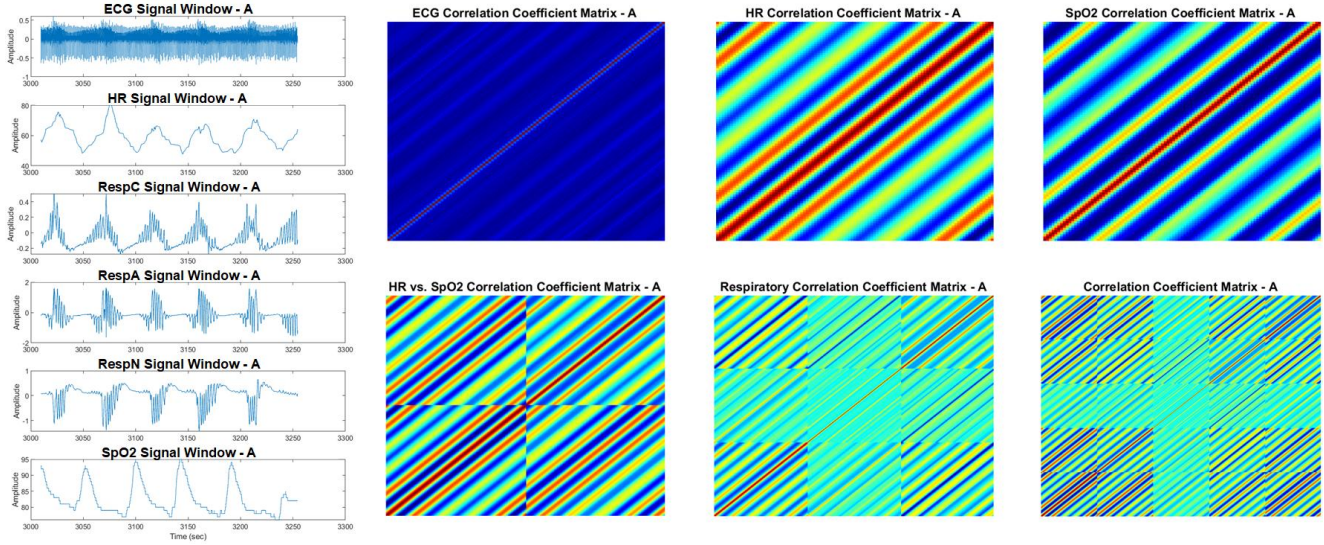


Figure 2. Apnea episode: time-domain signal windows and correlation coefficient matrices.

This cyclic pattern visualized in the physiological signals aligns with apnea and subsequent compensatory mechanisms reacting to apnea-induced oxygen desaturation during arousal and resumption of breathing. It determines clearly defined diagonals in the HR, respiratory signals and SpO_2 correlation coefficient matrices, respectively.

Following the prescribed procedure, the dataset for either signal space consists of 492 matrices: 316 for apnea episodes and 176 for normal breathing. Although the dataset contains only 492 samples, we claim that it is sufficient to test and validate the proposed apnea representation technique in CNN-based classification, especially in the context of transfer learning and data augmentation. Subsequently, the correlation coefficient matrices are saved as .csv files and converted into JPEG images using the JET color scheme.

III. CNN Architecture and Training

The classifier was developed in Matlab on the ResNet-50 architecture [15]. We have considered the pre-trained ResNet-50 model, originally trained on the ImageNet dataset, leveraging transfer learning to benefit the CNN from previously learned feature representations, which we apply to the proposed gradient-like image dataset. For adaptation to binary classification, the final ResNet50 fully connected layers have been replaced with a fully connected layer customized for binary classification, a softmax layer for probability distributions, and a classification layer to discriminate apnea from eupnea.

The image dataset is loaded, preserving the original RGB color scheme, but is resized to 224 x 224 pixels to comply with the ResNet-50 input dimensions. The dataset is split randomly into 80% training and 20% validation, ensuring a balanced distribution between apnea and eupnea records. We did not enforce subject-level separation, noting that this may result in subject overlap between the training and validation sets. Nevertheless, we employed 5-fold cross-validation across the entire dataset for a reliable model performance estimation.

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Due to the small dataset size, we have applied data augmentation techniques, such as random rotations (±30 degrees), translations (±10%), and scaling (80%-120%) uniformly to the training set, to expand dataset size, mitigate overfitting and enhance generalization of the CNN model. These augmentations magnitudes were kept low to preserve the spatial structure and inter-signal relationships within the correlation coefficient matrices, maintaining the physiological interpretability. For consistency during evaluation however, no augmentation was applied to the validation set.

Training was performed using the Adam optimizer with an initial learning rate of 3e-4 for 50 epochs. An early stopping criterion was implemented to break the training process if validation accuracy did not improve for 5 epochs to prevent overfitting.

The model performance is evaluated by means of accuracy (ACC), precision (Positive Predictive Value – PPV), sensitivity (True Positive Rate – TPR), specificity (True Negative Rate - TNR), F1 score, Matthews Correlation Coefficient (MCC) and Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) Score [16].

IV. Results

This study evaluates the classification performances of the proposed CNN-based procedure for sleep apnea detection.

The distribution of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) is plotted in the confusion matrices from Fig. 3 for each signal space. This enables a visualization of the model's classification performances to discriminate apnea events from normal breathing. The CNN performance metrics, determined from the confusion matrices for each signal space, are then listed in Table II.

TABLE II. CNN Performance Metrics

Dataset	CNN Performance Metrics						
	Acc [%]	*PPV* [%]	*TPR* [%]	*TNR* [%]	*F1* [%]	*MCC*	*ROC-AUC*
ECG	73.4	78.4	80.9	60	79.7	0.415	0.805
HR	69.3	76.2	76.2	57.1	76.2	0.333	0.805
SpO_2	80.6	81.4	90.5	62.8	85.7	0.565	0.823
HR+SpP2	89.7	89.5	95.2	80	92.3	0.775	0.970
Respiratory	96.9	95.4	100	91.4	97.6	0.934	0.990
All signals	97.9	96.9	100	94.3	98.4	0.955	0.996

ECG and HR signal spaces alone result in moderate classification performances. While both achieve weak to moderate accuracy values (73.4% and 69.3% respectively), they exhibit low specificity, i.e., high false positive rates, and moderate sensitivity, i.e., some positives are missed. The F1 scores (79,7% and 76.2 respectively) indicate a balance between precision and recall. However, the MCC scores (0.415 and 0.333 respectively) indicate a weak correlation to the actual labels, whereas the ROC-AUC scores (0.805 for both signal spaces) indicate that the model discriminates between apnea and normal cases with sub-optimal performance.

The SpO_2 signal space results in a higher accuracy (80.6%) than ECG and HR. Precision (81.4%) indicates a smaller false positives count, and recall (90.5%) suggests that most apnea cases are captured. However, specificity (62.8%) shows that some false positives persist. The F1-score (85.7%) indicates good balance between precision and recall, and MCC (0.565) and ROC-AUC (0.823) confirm the classification strength of the predictor.

The HR and SpO_2 combined signal space exhibits improved performance metrics compared to either individual signal space. Accuracy (89.7%) indicates a good detection of both apnea and normal cases. Precision (89.5%) indicates fewer false positives, recall (95.2%) indicates fewer false negatives, and specificity (80.0%) indicates the improvement in discriminating normal breathing. The F1-score (92.3%) confirms a strong balance between precision and recall, while MCC (0.775) and ROC-AUC (0.970) indicate near-optimal classification performance.

The respiratory signal space exhibits exceptional classification metrics. Accuracy (96.9%), precision (95.4%), i.e., small number of false positives, recall (100%), i.e., all apnea cases are detected with no false negatives, and specificity (91.4%), indicate that respiratory signals are the most reliable for apnea detection. The F1-score (97.6%) indicates optimal balance, the MCC (0.934) indicates correlation to true labels and the ROC-AUC (0.990) indicates ideal discrimination between classes.

Finally, the signal space consisting of all physiological signals, namely HR, SpO_2 and respiratory signals, provides the highest classification performances, with the accuracy reaching 97.9%, precision reaching 96.9%, recall reaching 100% and specificity reaching 98.4%. The F1-score (98.4%), MCC (0.955) and ROC-AUC (0.996) indicate that the employment of multiple physiological signals enhances classification performances and provides excellent reliability on the results.

The ROC curves for each signal space are plotted in Fig. 4, providing a comprehensive analysis on the model's capacity to discriminate between apnea episodes and normal breathing.

The CNN training process is illustrated on the signal space consisting of all physiological signals, by the accuracy and loss characteristics plotted in Fig. 5, for both training and validation datasets respectively. The remainder signal spaces exhibit similar training and validation trends. The training characteristics (solid line) indicate that both accuracy (plotted in blue) and loss (plotted in orange) improved rapidly over the first 50 iterations. After this training and validation accuracy stabilized above 90%, training loss continued to decrease, indicating that the model kept learning. Although validation loss shows some variation, the validation characteristics (dashed line) follow the trends in the training characteristics, indicating no signs of overfitting. Additionally, training stopped early after 18 epochs once the stopping criteria were met.

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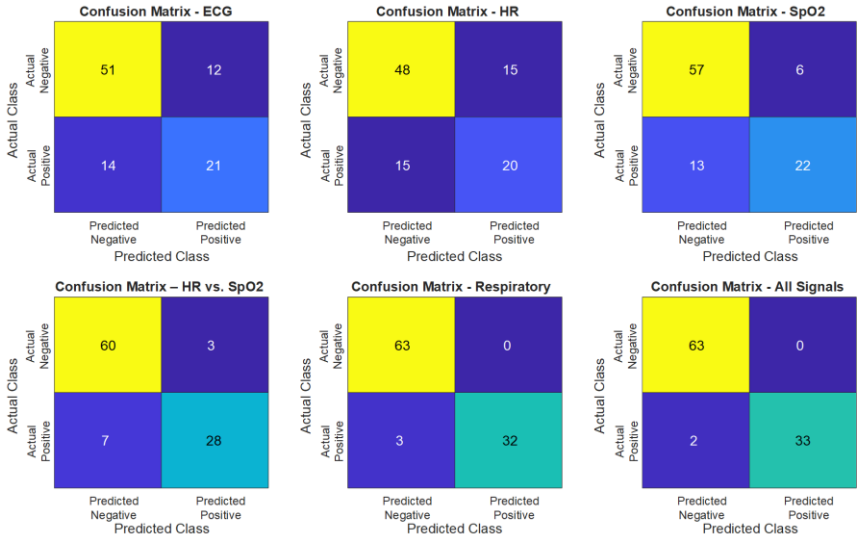


Figure 3. Confusion matrices for CNN-based apnea classification.

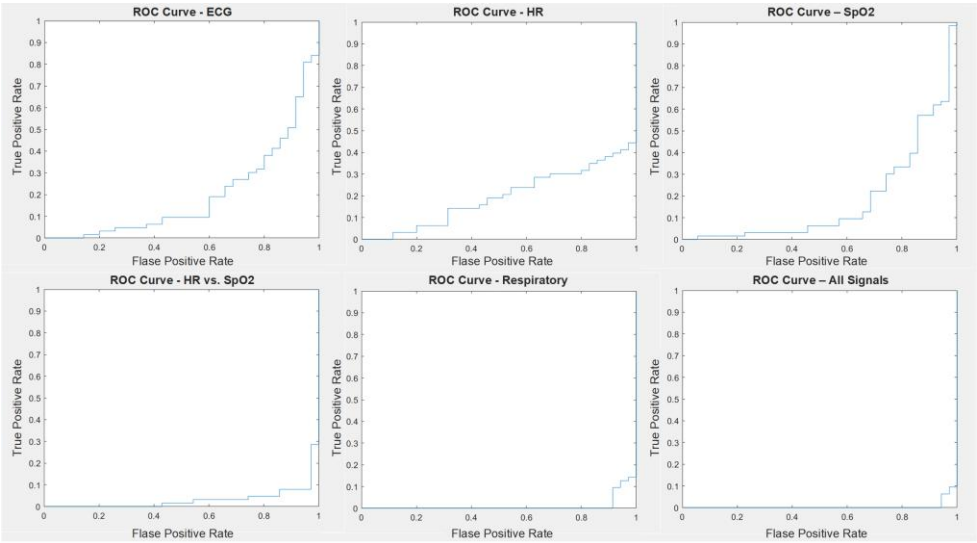


Figure 4. ROC curves for CNN-based apnea classification.

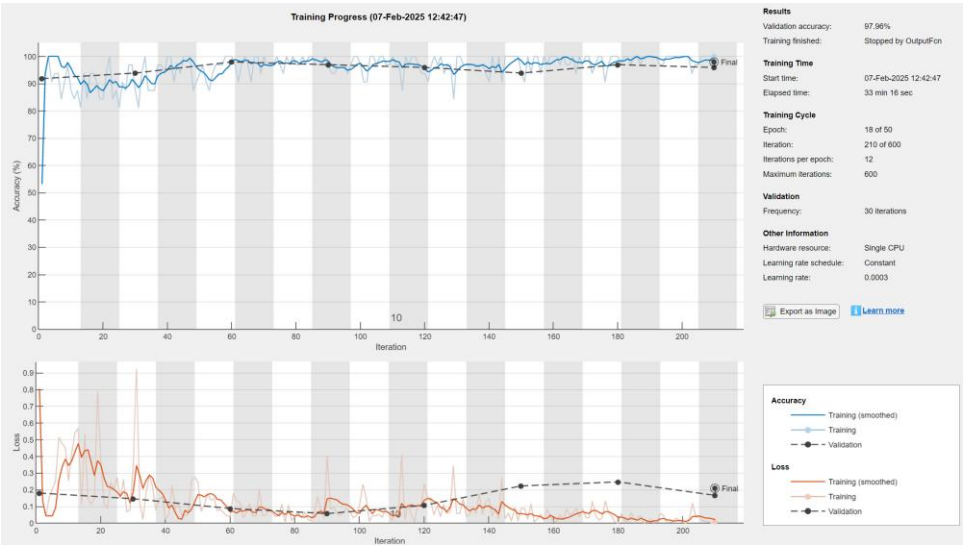


Figure 5. All signals dataset: training and validation accuracy and loss characteristics.

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V. Conclusion

Our research evaluates the benchmark ResNet-50 CNN model classification performances on a newly generated sleep apnea image dataset. The proposed representation technique evaluates the Pearson correlation coefficient on multiple physiological signal spaces. The purposed workflow is validated on recordings from the PhysioNet Apnea-ECG database, labeled "er" and consisting of ECG, respiratory signals and SpO_2.

Among the standalone physiological signal spaces, ECG, and the derived HR, presented limited predictive capability to detect apnea episodes, SpO_2 presented better performance metrics, while respiratory signals presented the best results proving the highest relevance for apnea discrimination.

The combined HR and SpO_2 signal space improved the classification performance metrics considerably. The best performances, namely 97.9% accuracy, an F1-score of 98.4%, an MCC of 0.955, and an AUC of 0.996, were achieved with the signal space that combines all physiological signals: HR, SpO_2 and respiration.

The performance metrics indicate that the largest contribution to classification performances with the combined dataset is brought by the respiratory signals and SpO_2. Nevertheless, we assert that HR also contains relevant information for apnea detection.

Given the high prevalence of apnea and its associated risks, early and accurate diagnosis is essential. While PSG is limited due to time and cost limitations, monitoring is more accessible in ambulatory settings or using portable monitoring devices which also enable remote monitoring.

In this context, the proposed correlation coefficient matrices provide a relevant representation of physiological signal interdependencies, allowing clinical insight into cardiorespiratory coordination during apnea episodes and enabling CNN-based classification. Clinically, this proposed workflow could be employed as a pre-screening tool in ambulatory or remote monitoring settings.

Although this study provided moderate performances on ECG-based classification, we hypothesize that ECG-derived features serve as primary indicators of apnea episodes. For future work, we plan to implement deep learning techniques for ECG-derived feature extraction and automatic apnea detection, with the goal of improving identification and diagnostic support in the context of real-time, non-invasive sleep apnea screening.

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